Introduction to Machine Learning
Practice Examination

Lectures 1–6 | Total: 100 marks

Course: ShanghaiTech CS182, 2026 Fall. This is an independent practice paper, not an official examination.

Name: _____ **Student ID:** _____

Date: _____ **Suggested time:** 150 minutes.

Scope: Introduction; Linear Algebra; Probability and Statistics; Optimization Primer; Bayesian Decision Theory; Parameter Estimation.

Section	Questions	Marks each	Total
A. True/False	20	1	20
B. Multiple response	20	2	40
C. Proofs and worked problems	8	5	40
Total	48		100

Scoring. Section A: 1 mark for the correct T/F response; 0 otherwise. Section B: each item has four options and exactly one, two, or three correct options. Select all options you believe are correct. A complete correct set earns 2 marks; a nonempty proper subset of the correct set earns 1 mark; a response containing any incorrect option, or an empty response, earns 0. There is no negative marking. Section C: show reasoning; subpart marks sum to 5 for each problem. Valid alternative proofs and methods receive equivalent credit. **Instructions.** Answer in English. T/F and multiple-response items need no explanation on the answer sheet. For Section C, state needed assumptions and show mathematical steps. Exact fractions, square roots, and natural logarithms are acceptable; a calculator is not required. All vectors are columns unless indicated otherwise. Read local assumptions and tie rules carefully. **Answer sheet — Section A (write T or F)**

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

Answer sheet — Section B (write option letters)

21	22	23	24	25	26	27	28	29	30

31	32	33	34	35	36	37	38	39	40

Section A. True/False 20 marks

Questions 1–10. Each question is worth 1 mark.

1. The model $h(x, w) = w_0 + w_1 \sin x + w_2 x^2$ is nonlinear in its parameter vector w. **[1 mark]**

2. For differentiable $f : \mathbb{R}^d \to \mathbb{R}$, using column-vector gradients, $\nabla_w f \in \mathbb{R}^d$ and $df = (\nabla_w f)^\mathsf{T} dw$ is a scalar. **[1 mark]**

3. If a hypothesis class has VC dimension four, it must shatter every set of four distinct points. **[1 mark]**

4. For a nonconstant i.i.d. normal sample, estimating both its unknown mean and variance by maximum likelihood uses divisor N, rather than $N - 1$, in the variance estimate. **[1 mark]**

5. The Lagrange dual function is concave in its multipliers even when the primal optimization problem is nonconvex. **[1 mark]**

6. For N independent and identically distributed observations with finite variance σ^2, the sample mean has variance σ^2/N. **[1 mark]**

7. For an absolutely continuous random variable, its probability density must be at most one at every point. **[1 mark]**

8. The posterior mean of a parameter is a Bayes-optimal point estimate under every possible loss function. **[1 mark]**

9. Under unit misclassification loss and class-independent rejection cost $0 < c < 1$, lowering c raises the posterior threshold $1 - c$ for accepting a class prediction. **[1 mark]**

10. Two mutually exclusive events, each having strictly positive probability, are independent. **[1 mark]**

Section A. True/False 20 marks

Questions 11–20. Each question is worth 1 mark.

11. If $A \in \mathbb{R}^{m \times n}$ has linearly independent columns, then $A^{\mathsf{T}}A$ is positive definite. **[1 mark]**

12. Every invertible real symmetric matrix is positive definite. **[1 mark]**

13. If two random variables have finite variances and zero covariance, they must be independent. **[1 mark]**

14. Features that are independent conditional on the class label can still be dependent when the class label is not observed. **[1 mark]**

15. For every i.i.d. sample of size $N \geq 2$ from a distribution with positive finite variance, $(\bar{X} - \mu)/(S/\sqrt{N})$ has an exact t_{N-1} distribution, where S^2 uses divisor $N - 1$. **[1 mark]**

16. A real orthogonal matrix Q preserves Euclidean lengths: $\|Qv\|_2 = \|v\|_2$ for every vector v of the appropriate dimension. **[1 mark]**

17. A hypothesis with zero training error is guaranteed to have zero error on future examples drawn from the same distribution. **[1 mark]**

18. If $f : \mathbb{R}^d \to \mathbb{R}$ is twice continuously differentiable and convex, then its Hessian is positive semidefinite at every point. **[1 mark]**

19. In a Lagrangian for a minimization problem with constraints $f_i(x) \leq 0$ and $h_j(x) = 0$, both inequality and equality multipliers must be nonnegative. **[1 mark]**

20. At a KKT point, every active inequality constraint must have a strictly positive multiplier. **[1 mark]**

Section B. Multiple Response 40 marks

Select all correct options. Each item has 1–3 correct options and is worth 2 marks.

21. There are two Gaussian classes in three dimensions. Count only free covariance parameters, excluding means and class priors. Which statements are correct?

(A) Two separate full covariance matrices require twelve parameters.

(B) A shared covariance of the form $s^2 I$ requires three parameters.

(C) One shared full covariance matrix requires six parameters.

(D) Separate diagonal covariance matrices require three parameters in total.

22. For $f(x_1, x_2) = x_1^2 + 2x_1x_2 + 4x_2^2$, which statements are correct?

(A) The Hessian has rank one.

(B) The Hessian is $\begin{bmatrix} 2 & 1 \\ 1 & 8 \end{bmatrix}$.

(C) $(0, 0)$ is the unique global minimizer.

(D) f is convex but not strictly convex.

23. Let $A \in \mathbb{R}^{2\times 3}$ and $v \in \mathbb{R}^2$ be fixed, and define $f(w) = v^\mathsf{T} A w$ for $w \in \mathbb{R}^3$. Which statements are correct?

(A) f is both convex and concave.

(B) $\nabla_w f = A^\mathsf{T} v$.

(C) $\nabla_w f = Av$.

(D) $\nabla_w^2 f$ is the zero 3×3 matrix.

24. Let $\mu = (0, 0)^\mathsf{T}$, $\Sigma = \mathrm{diag}(4, 1)$, and $x = (2, 1)^\mathsf{T}$. Define $d_M^2 = (x - \mu)^\mathsf{T}\Sigma^{-1}(x - \mu)$. Which statements are correct?

(A) The squared Euclidean distance from x to μ also equals two.

(B) $d_M^2 = 2$.

(C) The Mahalanobis distance d_M, rather than its square, equals two.

(D) The contour $z^\mathsf{T}\Sigma^{-1}z = 1$ has its longer semiaxis along the second coordinate.

25. Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. Which statements are correct?

(A) $\mathrm{rank}(A) = 2$.

(B) Full row rank of A makes $A^\mathsf{T}A$ invertible.

(C) $A^\mathsf{T}A$ is positive semidefinite and singular.

(D) $(-1, -1, 1)^\mathsf{T}$ lies in the null space of A.

Section B. Multiple Response 40 marks

Select all correct options. Each item has 1–3 correct options and is worth 2 marks.

26. Three classes have posterior probabilities $(0.46, 0.34, 0.20)$. Correct classification costs zero, an incorrect class prediction costs one, and rejection costs $c = 0.30$ for every true class. Which statements are correct?

(A) A class should be accepted whenever its posterior exceeds 0.30.

(B) With rejection available, rejection is the unique minimum-risk action.

(C) Without rejection, the minimum-error class prediction is C_1.

(D) The conditional risk of rejection is 0.54.

27. Let X_i be i.i.d. with mean μ and finite variance $0 < \sigma^2 < \infty$. Which statements are correct as $N \to \infty$?

(A) $\bar{X}_N$ is exactly normal for every finite N, even if the population is nonnormal.

(B) For every $\epsilon > 0$, $P(|\bar{X}_N - \mu| > \epsilon) \to 0$.

(C) The distribution of each individual observation X_i becomes normal merely because the sample size increases.

(D) $\sqrt{N}(\bar{X}_N - \mu)/\sigma$ converges in distribution to $\mathcal{N}(0, 1)$.

28. A fixed training set is fitted by unregularized least squares over nested polynomial classes of degrees M and $M + 1$. Both global minima are attained. Which statements are correct?

(A) The degree with the smallest training error necessarily also has the smallest expected test error.

(B) The minimum training sum of squares must strictly decrease whenever one basis function is added.

(C) An independent validation set can be used to select the polynomial degree.

(D) The minimum training sum of squares cannot increase when the degree rises from M to $M + 1$.

29. Let X, Y have finite, strictly positive variances. No independence assumption is made unless explicitly stated in an option. Which statements are correct?

(A) If X and Y are independent, then $\mathrm{Corr}(X, Y) = 0$.

(B) $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ always holds.

(C) $E[X + Y] = E[X] + E[Y]$.

(D) $\mathrm{Var}(3X - 2) = 9\,\mathrm{Var}(X)$.

30. For $f(x) = 2(x - b)^2$, gradient descent uses a constant step size $\eta > 0$ and starts at $x_0 \neq b$. Which statements are correct?

(A) Every positive step size yields convergence because f is strictly convex.

(B) With $\eta = 3/5$, the magnitude of $x_t - b$ grows geometrically.

(C) With $\eta = 1/2$, the iterates alternate and do not converge to b.

(D) With $\eta = 1/4$, the first update reaches b.

Section B. Multiple Response 40 marks

Select all correct options. Each item has 1–3 correct options and is worth 2 marks.

31. Two classes have $P(C_1) = 0.2$, $P(C_2) = 0.8$. An event E satisfies $P(E \mid C_1) = 0.6$ and $P(E \mid C_2) = 0.1$. Which statements are correct?

(A) $P(C_1 \mid E) = 0.6$.

(B) After observing E, the minimum conditional error under 0–1 loss is 0.2.

(C) The posterior odds $P(C_1 \mid E)/P(C_2 \mid E)$ equal six.

(D) $P(E) = 0.7$.

32. Let the observed dataset be $\mathcal{D}$, with positive likelihood $L(\theta) = p(\mathcal{D} \mid \theta)$. Which statements are correct?

(A) Maximizing $\log L(\theta)$ preserves the maximizers of $L(\theta)$.

(B) $\int L(\theta)\, d\theta = 1$ is required because likelihood is a density over parameters.

(C) A constant positive prior on a parameter domain makes MAP and MLE coincide when optimized over that same domain.

(D) Under conditional independence of a future observation and $\mathcal{D}$ given θ, $p(x_{\text{new}} \mid \mathcal{D}) = \int p(x_{\text{new}} \mid \theta) p(\theta \mid \mathcal{D})\, d\theta$.

33. Let X_i be i.i.d. $\mathcal{N}(\mu, \sigma^2)$, and estimate μ by $T = \bar{X}/2$. Which statements are correct?

(A) T is unbiased for all values of μ.

(B) $\mathrm{Var}(T) = \sigma^2/(4N)$.

(C) $E[(T - \mu)^2] = \sigma^2/(4N) + \mu^2/4$.

(D) The bias of T is $-\mu/2$.

34. Consider a minimization problem with a finite primal optimum and inequality constraints written as $f_i(x) \le 0$. Let $g(\lambda) = \inf_x [f_0(x) + \sum_i \lambda_i f_i(x)]$. Which statements are correct?

(A) $g(\lambda)$ can equal $-\infty$ for some nonnegative λ.

(B) Weak duality requires the optimal dual value to be at least the optimal primal value.

(C) Zero optimal duality gap, by itself, guarantees that the primal infimum is attained.

(D) g is concave in λ, whether or not the primal problem is convex.

35. Binary variables X, Y have joint masses $p(0,0) = 0.1$, $p(0,1) = 0.2$, $p(1,0) = 0.3$, and $p(1,1) = 0.4$. Which statements are correct?

(A) $\mathrm{Cov}(X, Y) = -0.02$.

(B) $P(Y = 1) = 0.6$.

(C) $P(Y = 1 \mid X = 1) = 2/3$.

(D) X and Y are independent.

Section B. Multiple Response 40 marks

Select all correct options. Each item has 1–3 correct options and is worth 2 marks.

36. Let $X_1, \dots, X_N$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$, with $N \geq 2$ and $\sigma^2 > 0$. Put $S^2 = \frac{1}{N-1}\sum_i (X_i - \bar{X})^2$. Which statements are correct?

(A) $(\bar{X} - \mu)/(S/\sqrt{N}) \sim t_N$.

(B) $\bar{X}$ and S^2 are independent.

(C) $(N-1)S^2/\sigma^2 \sim \chi^2_N$.

(D) $(N-1)S^2/\sigma^2 \sim \chi^2_{N-1}$.

37. For two equally likely classes, conditionally independent binary features have success probabilities $(p_{11}, p_{12}) = (0.8, 0.3)$ and $(p_{21}, p_{22}) = (0.4, 0.7)$. Observe $x = (1, 0)$. Which statements are correct?

(A) The unnormalized log-posterior score for C_1 is $\log(0.28)$.

(B) The zero-valued second feature contributes $1 - p_{i2}$ to the likelihood of class C_i.

(C) $P(C_1 \mid x) = 0.56$.

(D) $p(x \mid C_2) = 0.4 + 0.3 = 0.7$.

38. An i.i.d. Bernoulli sample $X_1, \dots, X_N$ has parameter $\theta \in [0, 1]$. Consider the estimator $\widehat{\theta} = \bar{X}$. Which statements are correct?

(A) $\mathrm{Var}_\theta(\widehat{\theta}) = \theta(1-\theta)/N$.

(B) If every observed value equals one, the likelihood is maximized at $\theta = 1$.

(C) For every $\theta \in [0, 1]$, $E_\theta[\widehat{\theta}] = \theta$.

(D) When all observations are one, no MLE exists because the likelihood has no interior stationary point.

39. Minimize x^2 subject to $1 - x \leq 0$, with Lagrangian $L(x, \lambda) = x^2 + \lambda(1 - x)$. Which statements are correct?

(A) The objective gradient must be zero at every constrained optimum.

(B) $x^* = 1$ and $\lambda^* = 0$ satisfy stationarity.

(C) $x^* = 0$ and $\lambda^* = 0$ solve the constrained problem.

(D) $x^* = 1$ and $\lambda^* = 2$ satisfy all KKT conditions.

40. At an observation x, the class posterior is $(P(C_1 \mid x), P(C_2 \mid x)) = (0.55, 0.45)$. Losses are $\lambda_{11} = \lambda_{22} = 0$, $\lambda_{12} = 4$, and $\lambda_{21} = 1$, with rows denoting actions. Which statements are correct?

(A) The minimum conditional risk equals the minimum conditional error probability, 0.45.

(B) $R(\alpha_1 \mid x) = 4(0.55) = 2.2$.

(C) Choosing C_2 minimizes conditional risk.

(D) Choosing C_1 minimizes conditional risk because its posterior is larger.

Section C. Proofs and Worked Problems

41. Quadratic forms and differentials **[5 marks]**

Mathematical proof

Let $A \in \mathbb{R}^{d\times d}$ be fixed, not necessarily symmetric, and let $f(w) = \frac{1}{2} w^{\mathsf{T}} A w$ for a column vector $w \in \mathbb{R}^d$.

(a) Prove that $w^{\mathsf{T}} A w = w^{\mathsf{T}}[(A + A^{\mathsf{T}})/2]w$. Your proof must account for the skew-symmetric part of A. **[2 marks]**

(b) Using the differential product rule and $df = (\nabla_w f)^{\mathsf{T}} dw$, prove $\nabla_w f = \frac{1}{2}(A + A^{\mathsf{T}})w$. Explain exactly when this simplifies to Aw. **[3 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

42. Maximum of affine functions **[5 marks]**

Mathematical proof

For fixed $a_i \in \mathbb{R}^d$, $b_i \in \mathbb{R}$, and $m \geq 1$, define $F(x) = \max_{1 \leq i \leq m}(a_i^\top x + b_i)$.

(a) Prove directly from the definition that F is convex on $\mathbb{R}^d$. Do not assume it is differentiable. **[3 marks]**

(b) Take $d = 1$, $m = 2$, $a_1 = 1$, $a_2 = -1$, and $b_1 = b_2 = 0$. Prove that this F is not differentiable at zero and is not strictly convex. State what this shows about using an ordinary Hessian to establish convexity. **[2 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

43. A correlation bound and its equality case **[5 marks]**

Mathematical proof

Let X, Y have finite, strictly positive variances. Define $U = X - E[X]$, $V = Y - E[Y]$, and $\rho = \operatorname{Cov}(X, Y)/\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}$.

(a) Prove $|\rho| \le 1$ by using $E[(U - tV)^2] \ge 0$ for every real t. **[3 marks]**

(b) Prove that $|\rho| = 1$ holds if and only if $U = cV$ almost surely for some nonzero real constant c. State how the sign of c determines ρ. **[2 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

44. A certificate from weak duality **[5 marks]**

Mathematical proof

Consider $p^* = \inf\{f_0(x) : f_i(x) \le 0,\ h_j(x) = 0\}$, assumed finite. For $\lambda_i \ge 0$ and arbitrary real ν_j, let $g(\lambda, \nu) = \inf_z [f_0(z) + \sum_i \lambda_i f_i(z) + \sum_j \nu_j h_j(z)]$. Assume $g(\lambda, \nu) > -\infty$.

(a) Prove $g(\lambda, \nu) \le p^* \le f_0(x)$ for every feasible x. Your proof must not assume that the primal infimum is attained or that strong duality holds. **[3 marks]**

(b) For $\epsilon \ge 0$, prove that a feasible x satisfying $f_0(x) - g(\lambda, \nu) \le \epsilon$ has $0 \le f_0(x) - p^* \le \epsilon$. Explain why equality multipliers need no sign restriction in this argument. **[2 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

45. Three classes, unequal losses, and rejection **[5 marks]**

Calculation and decision

At a fixed observation x, the priors are $(0.2, 0.5, 0.3)$ and the class-conditional density values are $(0.6, 0.2, 0.4)$. The loss matrix is

$$\Lambda = (\lambda_{ik}) = \begin{bmatrix} 0 & 2 & 4 \\ 3 & 0 & 1 \\ 1 & 5 & 0 \end{bmatrix},$$

where row i denotes prediction C_i and column k denotes the true class. Rejection costs $c = 6/5$ for every true class. Break any risk tie in favor of rejection.

(a) Compute the evidence and all three posterior probabilities. **[2 marks]**

(b) Compute each class-action conditional risk and the rejection risk. Identify the optimal action. **[2 marks]**

(c) Now change only the rejection cost to $c = 7/5$. Identify the optimal action and explain why a rule based only on the largest posterior is insufficient here. **[1 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

46. Normal observations with a normal prior **[5 marks]**

Derivation and prediction

Conditionally on θ, the observations $X_1, \dots, X_N$ are independent $\mathcal{N}(\theta, \sigma^2)$, where $\sigma^2 > 0$ is known. The prior is $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$, with $\tau_0^2 > 0$.

(a) Starting from likelihood times prior and completing the square, derive the posterior mean m_N and posterior variance v_N. Do not merely quote a conjugacy formula. **[3 marks]**

(b) For $N = 3$, observations $(2, 4, 6)$, $\sigma^2 = 4$, $\mu_0 = 1$, and $\tau_0^2 = 2$, calculate m_N, v_N, and the MAP estimate. **[1 marks]**

(c) A new observation is conditionally independent of the training data given θ, with $X_{\text{new}} \mid \theta \sim \mathcal{N}(\theta, 4)$. Compute its posterior predictive mean and variance for the numerical case. Explain the additional variance beyond four. **[1 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

47. A Gaussian decision boundary **[5 marks]**

Derivation and calculation

Under 0–1 loss, two classes have means $\mu_1 = (0, 0)^\mathsf{T}$ and $\mu_2 = (2, 1)^\mathsf{T}$, shared covariance $\Sigma = \mathrm{diag}(2, 1)$, and priors $P(C_1) = 1/4$, $P(C_2) = 3/4$. Use $g_i(x) = \log p(x \mid C_i) + \log P(C_i)$. Break score ties in favor of C_2.

(a) Derive an explicit affine expression for $g_2(x) - g_1(x)$, and give the decision boundary and class regions. **[3 marks]**

(b) Classify $x_A = (0, 0)^\mathsf{T}$ and $x_B = (1, 0)^\mathsf{T}$. You may use $1 < \log 3 < 3/2$. **[1 marks]**

(c) If the covariance is changed to two separate full 2×2 covariance matrices, how many free covariance parameters are needed? Compare with the shared full-covariance case. **[1 marks]**

Work space (continue on separate paper if needed).

Section C. Proofs and Worked Problems

48. A constrained quadratic and its dual **[5 marks]**

Derivation and calculation

Solve

$$\min_{u,v\in\mathbb{R}} \ \frac{1}{2}[(u-2)^2+(v+1)^2] \quad \text{subject to} \quad u+v=0, \quad u\le 1.$$

Use $L(u,v,\lambda,\nu)=\frac{1}{2}[(u-2)^2+(v+1)^2]+\lambda(u-1)+\nu(u+v)$, with $\lambda\ge 0$ and $\nu\in\mathbb{R}$.

(a) Find the primal optimizer and objective value. Establish that it is the unique global optimizer. **[2 marks]**

(b) Find multipliers and verify all KKT conditions explicitly. **[2 marks]**

(c) Find $g(\lambda,\nu)=\inf_{u,v} L(u,v,\lambda,\nu)$, evaluate it at your multipliers, and give the resulting duality gap. **[1 marks]**

Work space (continue on separate paper if needed).